Year 10 Mathematics

Algebra 1

2.1 Substitution

Substitution

STARTER · 5 MIN

Date: _____ **Title:** Substitution

LEARNING OBJECTIVES

- Substitute positive and negative numbers into expressions
- Substitute into expressions with powers and brackets
- Substitute into formulae and solve problems

STARTER *Answer in your book. No calculators.*

1. Work out $-3 + 7$

2. Work out -4×-5

3. Work out $(-3)^2$

4. Work out $2 \times 5 - 3 \times 4$

Starter — Answers

STARTER

Tick or correct in green pen

1. $-3 + 7 =$ **4**

2. $-4 \times (-5) =$ **20**

3. $(-3)^2 = (-3) \times (-3) =$ **9**

4. $2 \times 5 - 3 \times 4 = 10 - 12 =$ **−2**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute positive and negative numbers into expressions	☐
SILVER	I can substitute into expressions with powers and brackets	☐
GOLD	I can substitute into formulae and solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Substitution

INTRODUCTION · 3 MIN

Substitute means replace each letter with its number. Put negative numbers in brackets.

Replace

Swap each letter for its value.

3a when a = 5: 3 × 5 = 15

Brackets

Put negative numbers in brackets.

x^2 when x = −2: $(-2)^2 = 4$

Order

Use BIDMAS: powers, then × and ÷, then + and −.

2 + 3 × 4 = 2 + 12 = 14

Formula

Substitute every letter, then work it out.

v = u + at, u = 2, a = 3, t = 4: v = 14

Examples 1 and 2 — Substituting Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

$a = 4$ and $b = 3$

Find the value of $5a - 2b$

Example 2

$x = -2$

Find the value of $3x + 10$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$5a - 2b = 5 \times 4 - 2 \times 3$$
$$= 20 - 6$$
$$\mathbf{= 14}$$

Tip: write the numbers in first, then work it out.

Example 2

$$3x + 10 = 3 \times (-2) + 10$$
$$= -6 + 10$$
$$\mathbf{= 4}$$

Tip: put negative numbers in brackets.

Task 1

PRACTICE · 9 MIN

BRONZE

a = 5, b = 2, c = −3. Find the value of each expression.

1. a + b

2. 3a − b

3. 4c

4. ab + c

5. a − 2c

★ C = 4h + 5. Find C when h = 3

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $a + b = 5 + 2 =$ **7**

2. $3a - b = 15 - 2 =$ **13**

3. $4c = 4 \times (-3) =$ **−12**

4. $ab + c = 10 + (-3) =$ **7**

5. $a - 2c = 5 - (-6) =$ **11**

★ $C = 4 \times 3 + 5 =$ **17**

Examples 3 and 4 — Powers and Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

$x = 3$ and $y = -2$

Find the value of $x^2 + y^2$

Example 4

$p = 4$ and $q = -1$

Find the value of $3(p - q)^2$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x^2 + y^2 = 3^2 + (-2)^2$$
$$= 9 + 4$$
$$= \mathbf{13}$$

Tip: $(-2)^2 = (-2) \times (-2) = +4$

Example 4

$$3(p - q)^2 = 3(4 - (-1))^2$$
$$= 3 \times 5^2$$
$$= \mathbf{3 \times 25 = 75}$$

Tip: work out the bracket first, then square, then multiply.

Task 2

PRACTICE · 9 MIN

SILVER

$x = 4$, $y = -3$. Find the value of each expression.

1. x^2

2. y^2

3. $2x^2$

4. $(x + y)^2$

5. $\dfrac{x^2 + y^2}{5}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x^2 = 4^2 =$ **16**

2 $y^2 = (-3)^2 =$ **9**

3 $2x^2 = 2 \times 16 =$ **32**

4 $(x + y)^2 = (4 + (-3))^2 =$ **$1^2 = 1$**

5 $$\frac{x^2 + y^2}{5} = \frac{16 + 9}{5} = \frac{25}{5} =$$ **5**

Examples 5 and 6 — Formulae

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

$v = u + at$

Work out the value of v when u = 12, a = −3 and t = 5

Example 6 (2 marks)

$$E = \frac{1}{2}mv^2$$

Work out the value of E when m = 6 and v = −4

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$v = 12 + (-3) \times 5$$
$$= 12 - 15$$
$$\mathbf{v = -3}$$

Tip: multiply before you add.

Example 6

$$E = \frac{1}{2} \times 6 \times (-4)^2$$
$$= 3 \times 16$$
$$\mathbf{E = 48}$$

Tip: square v first: $(-4)^2 = 16$

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. $s = ut + \frac{1}{2}at^2$. Work out s when u = 3, t = 4 and a = 10 (2 marks)

2. P = 2l + 2w. Work out P when l = 7.5 and w = 4.2 (2 marks)

3. $T = \frac{a + b}{c}$ (2 marks)

   Work out T when a = 12, b = −4 and c = −2

4. $y = 3x^2 - 2x$. Work out y when x = −2 (2 marks)

5. A taxi charges C = 2.5 + 1.8m pounds for a journey of m km. Work out the cost of an 8 km journey. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $s = 3 \times 4 + \frac{1}{2} \times 10 \times 4^2 = 12 + 80 =$ **92**

2. $P = 15 + 8.4 =$ **23.4**

3. $T = \dfrac{12 + (-4)}{-2} =$ **−4**

4. $y = 3 \times 4 - 2 \times (-2) = 12 + 4 =$ **16**

5. $C = 2.5 + 1.8 \times 8 = 2.5 + 14.4 =$ **£16.90**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $a = 6$ and $b = -2$. Find $2a + 3b$

SILVER $x = -3$. Find $2x^2 - x$

GOLD $F = \frac{9C}{5} + 32$

Work out F when $C = -10$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $2a + 3b = 2 \times 6 + 3 \times (-2) = 12 - 6 =$ **6**

SILVER $2x^2 - x = 2 \times (-3)^2 - (-3) = 18 + 3 =$ **21**

GOLD $F = \dfrac{9 \times (-10)}{5} + 32 = -18 + 32 =$ **14**

Exam Practice — Substituting Numbers

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

a = 5 and b = −3

(a) Work out the value of 4a + 2b (1)

.....

(b) Work out the value of ab (1)

.....

Question 2

(2 marks)

The diagram shows a rectangle.

Work out the perimeter when x = 4.

Diagram NOT accurately drawn



..... cm

Exam Practice — Powers and Formulae

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

$x = -2$ and $y = 3$

(a) Work out the value of $x^2 + y^2$ (1)

.....

(b) Work out the value of $2(x - y)^2$ (1)

.....

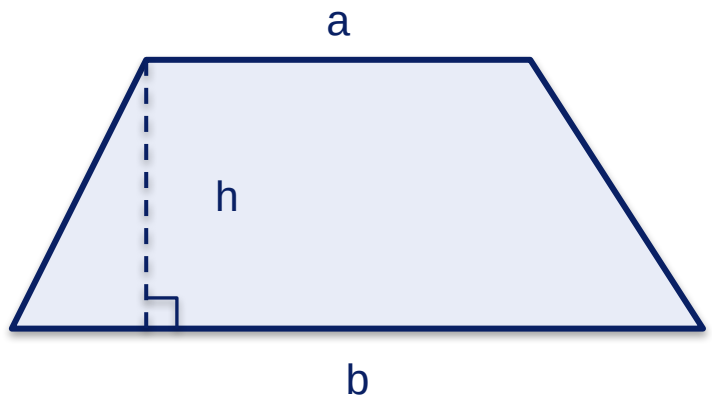
Question 4

(2 marks)

Area of a trapezium: $A = \frac{1}{2}(a + b)h$

Work out A when a = 3.5, b = 6.5, h = 4.

Diagram NOT accurately drawn



.....

Exam Practice — Formulae in Context

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

$v^2 = u^2 + 2as$

Work out the value of v when u = 6, a = −2.5 and s = 4.

v is positive.

v = ...

Question 6

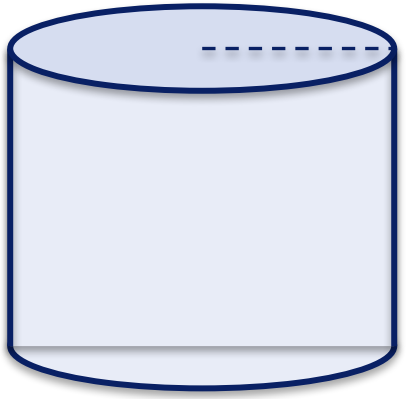
(2 marks)

The volume of a cylinder is $V = \pi r^2 h$

Work out V when r = 3.5 cm and h = 10 cm.

Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... cm^3

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 20 + (−6) = 14 — B1

(b) 5 × −3 = −15 — B1

Tip: put negative numbers in brackets.

Question 2

length = 2 × 4 + 3 = 11 — M1

11 + 4 + 11 + 4 = 30 cm — A1

Tip: perimeter = add all four sides.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 4 + 9 = 13 B1

(b) $2 \times (-5)^2 = 2 \times 25$

= 50 B1

Tip: $(-2)^2 = +4$, a square is never negative.

Question 4

$\frac{1}{2} \times (3.5 + 6.5) \times 4$ M1

= 20 A1

Tip: add inside the bracket first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$v^2 = 36 + 2 \times (-2.5) \times 4$ **M1**

$v^2 = 36 - 20 = 16$

$v = 4$ **A1**

Tip: find v^2 first, then square root.

Question 6

$\pi \times 3.5^2 \times 10$ **M1**

$= 384.84...$

$= 385\ cm^3$ **A1**

Tip: square r before multiplying by π.